

# Two derived quadratic equations from a real quadratic equation with two distinct positive roots

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## Abstract

Let

$$ax^2 + bx + c = 0, \quad a \neq 0$$

be a real quadratic equation with two distinct positive roots

$$\alpha > \beta > 0.$$

For real parameters  $m, n, p, q$ , this paper derives, in closed form, the quadratic equations whose roots are

$$m\alpha^p, \quad n\beta^q$$

and

$$n\alpha^q, \quad m\beta^p.$$

The derivation proceeds through Viète's formulae, symmetric root parametrisation, logarithmic power transformations, and monic quadratic reconstruction. We also present algorithms, statistical validation methods, numerical stability considerations, and an AI/ML interpretation of the transformation as a differentiable feature map on the space of positive-root quadratics.

The paper ends with “The End”

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## 1 Introduction

A quadratic equation

$$ax^2 + bx + c = 0, \quad a \neq 0,$$

with roots  $\alpha, \beta$ , satisfies Viète's formulae

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

When the roots are ordered and positive,

$$\alpha > \beta > 0,$$

real powers such as  $\alpha^p$  and  $\beta^q$  are unambiguous for all real  $p, q$ , because

$$\alpha^p = \exp(p \log \alpha), \quad \beta^q = \exp(q \log \beta).$$

The main problem is to find, using only the data

$$a, b, c, m, n, p, q,$$

the two quadratic equations with roots

$$m\alpha^p, \quad n\beta^q$$

and

$$n\alpha^q, \quad m\beta^p.$$

The central method is:

1. recover  $\alpha, \beta$  symmetrically from the coefficients;
2. express  $\log \alpha, \log \beta$  using symmetric logarithmic coordinates;
3. apply the exponent transformations;
4. reconstruct the target quadratic by its root sum and root product.

## 2 Assumptions and Notation

Let

$$Q(x) = ax^2 + bx + c, \quad a \neq 0.$$

Its discriminant is

$$\Delta = b^2 - 4ac.$$

The assumption that  $Q$  has two distinct positive roots is equivalent to

$$\Delta > 0, \quad -\frac{b}{a} > 0, \quad \frac{c}{a} > 0.$$

Define the symmetric coefficient quantities

$$S = -\frac{b}{a}, \quad P = \frac{c}{a}, \quad R = \sqrt{S^2 - 4P}.$$

Then

$$S = \alpha + \beta, \quad P = \alpha\beta, \quad R = \alpha - \beta,$$

because  $\alpha > \beta$ . Hence

$$\alpha = \frac{S + R}{2}, \quad \beta = \frac{S - R}{2}.$$

For logarithmic transformations, define

$$L = \frac{1}{2} \log P,$$

and

$$T = \frac{1}{2} \log \left( \frac{S + R}{S - R} \right).$$

Since  $\alpha, \beta > 0$ , both logarithms are well-defined. Moreover,

$$L = \frac{1}{2} (\log \alpha + \log \beta),$$

and

$$T = \frac{1}{2} (\log \alpha - \log \beta).$$

Therefore

$$\log \alpha = L + T, \quad \log \beta = L - T,$$

and hence

$$\alpha = e^{L+T}, \quad \beta = e^{L-T}.$$

| Symbol   | Meaning        | Formula                             |
|----------|----------------|-------------------------------------|
| $S$      | Root sum       | $-b/a$                              |
| $P$      | Root product   | $c/a$                               |
| $R$      | Root gap       | $\sqrt{S^2 - 4P}$                   |
| $L$      | Mean log-root  | $\frac{1}{2} \log P$                |
| $T$      | Half log-ratio | $\frac{1}{2} \log((S + R)/(S - R))$ |
| $\alpha$ | Larger root    | $e^{L+T} = (S + R)/2$               |
| $\beta$  | Smaller root   | $e^{L-T} = (S - R)/2$               |

Table 1: Core quantities used throughout the paper.

### 3 Geometric Interpretation

The roots  $\alpha, \beta$  can be represented by their arithmetic centre and gap:

$$\frac{\alpha + \beta}{2} = \frac{S}{2}, \quad \frac{\alpha - \beta}{2} = \frac{R}{2}.$$

The logarithmic representation instead uses the geometric centre and log-gap:

$$\sqrt{\alpha\beta} = e^L, \quad \sqrt{\frac{\alpha}{\beta}} = e^T.$$

Thus,

$$\alpha = e^L e^T, \quad \beta = e^L e^{-T}.$$

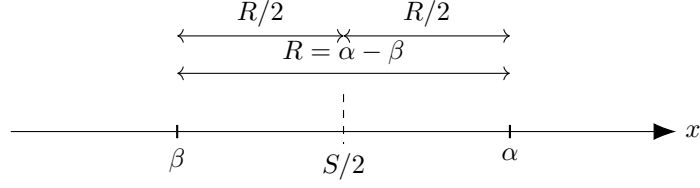


Figure 1: Arithmetic parametrisation of two ordered positive roots.

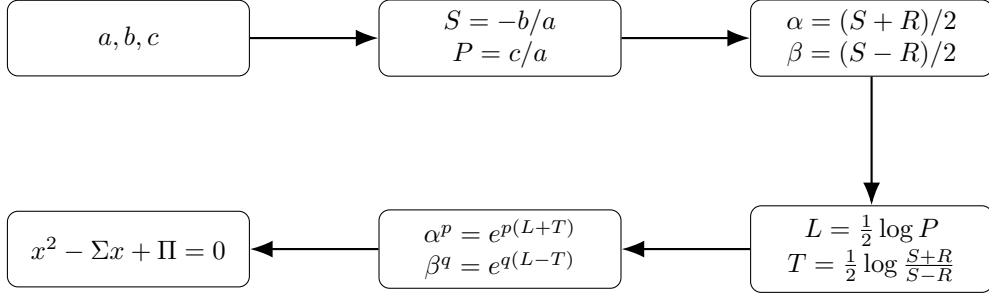


Figure 2: Transformation pipeline from coefficients to transformed quadratics.

## 4 Stage 1: Linear Scaling of Ordered Roots

We first solve the simpler problem of transforming

$$\alpha, \beta$$

into

$$m\alpha, n\beta$$

and

$$n\alpha, m\beta.$$

**Theorem 1** (Linear Scaling Transformation). *Let*

$$gx^2 + hx + i = 0$$

*have roots  $\alpha > \beta > 0$ . Define*

$$S_g = -\frac{h}{g}, \quad P_g = \frac{i}{g}, \quad R_g = \sqrt{S_g^2 - 4P_g}.$$

*Then the quadratic with roots*

$$m\alpha, \quad n\beta$$

*is*

$$x^2 - \frac{(m+n)S_g + (m-n)R_g}{2}x + mnP_g = 0.$$

*The quadratic with roots*

$$n\alpha, \quad m\beta$$

*is*

$$x^2 - \frac{(m+n)S_g - (m-n)R_g}{2}x + mnP_g = 0.$$

*Proof.* Since

$$\alpha = \frac{S_g + R_g}{2}, \quad \beta = \frac{S_g - R_g}{2},$$

we have

$$m\alpha + n\beta = m\frac{S_g + R_g}{2} + n\frac{S_g - R_g}{2}.$$

Therefore

$$m\alpha + n\beta = \frac{(m+n)S_g + (m-n)R_g}{2}.$$

Also,

$$(m\alpha)(n\beta) = mn\alpha\beta = mnP_g.$$

Hence the quadratic with roots  $m\alpha, n\beta$  is

$$x^2 - (m\alpha + n\beta)x + (m\alpha)(n\beta) = 0,$$

which gives the first formula.

Similarly,

$$n\alpha + m\beta = n\frac{S_g + R_g}{2} + m\frac{S_g - R_g}{2} = \frac{(m+n)S_g - (m-n)R_g}{2},$$

and

$$(n\alpha)(m\beta) = mnP_g.$$

Thus the second formula follows. □

## 5 Stage 2: Power Transformation of Positive Roots

We now transform the roots into powers:

$$\alpha^p, \beta^q$$

and

$$\alpha^q, \beta^p.$$

**Theorem 2** (Power Transformation). *Let*

$$dx^2 + ex + f = 0$$

*have roots  $\alpha > \beta > 0$ . Define*

$$S_d = -\frac{e}{d}, \quad P_d = \frac{f}{d}, \quad R_d = \sqrt{S_d^2 - 4P_d},$$

$$L_d = \frac{1}{2} \log P_d,$$

*and*

$$T_d = \frac{1}{2} \log \left( \frac{S_d + R_d}{S_d - R_d} \right).$$

*Then the quadratic with roots*

$$\alpha^p, \beta^q$$

*is*

$$x^2 - \left( e^{p(L_d+T_d)} + e^{q(L_d-T_d)} \right) x + e^{(p+q)L_d+(p-q)T_d} = 0.$$

*The quadratic with roots*

$$\alpha^q, \beta^p$$

*is*

$$x^2 - \left( e^{q(L_d+T_d)} + e^{p(L_d-T_d)} \right) x + e^{(p+q)L_d+(q-p)T_d} = 0.$$

*Proof.* Because  $\alpha, \beta > 0$ , real powers can be written using natural logarithms:

$$\alpha^p = e^{p \log \alpha}, \quad \beta^q = e^{q \log \beta}.$$

By construction,

$$\log \alpha = L_d + T_d, \quad \log \beta = L_d - T_d.$$

Thus

$$\alpha^p = e^{p(L_d+T_d)}, \quad \beta^q = e^{q(L_d-T_d)}.$$

The sum of the transformed roots is

$$e^{p(L_d+T_d)} + e^{q(L_d-T_d)}.$$

The product is

$$e^{p(L_d+T_d)} e^{q(L_d-T_d)} = e^{(p+q)L_d+(p-q)T_d}.$$

Reconstructing the monic quadratic gives the first formula.

The second formula follows by replacing  $p, q$  with  $q, p$  in the corresponding root positions. □

## 6 Main Result

We now combine the preceding two stages.

**Theorem 3** (The  $m\alpha^p, n\beta^q$  Quadratics). *Let*

$$ax^2 + bx + c = 0$$

*have roots*

$$\alpha > \beta > 0.$$

*Let  $m, n, p, q \in \mathbb{R}$ . Define*

$$S = -\frac{b}{a}, \quad P = \frac{c}{a}, \quad R = \sqrt{S^2 - 4P},$$

$$L = \frac{1}{2} \log P,$$

*and*

$$T = \frac{1}{2} \log \left( \frac{S + R}{S - R} \right).$$

*Then the quadratic with roots*

$$m\alpha^p, \quad n\beta^q$$

*is*

$$x^2 - \left( me^{p(L+T)} + ne^{q(L-T)} \right) x + mne^{(p+q)L+(p-q)T} = 0.$$

*The quadratic with roots*

$$n\alpha^q, \quad m\beta^p$$

*is*

$$x^2 - \left( ne^{q(L+T)} + me^{p(L-T)} \right) x + mne^{(p+q)L+(q-p)T} = 0.$$

*Proof.* By Viète's formulae,

$$\alpha + \beta = S, \quad \alpha\beta = P.$$

Since  $\alpha > \beta$ ,

$$R = \sqrt{S^2 - 4P} = \alpha - \beta.$$

Therefore

$$\alpha = \frac{S + R}{2}, \quad \beta = \frac{S - R}{2}.$$

The logarithmic coordinates satisfy

$$\log \alpha = L + T, \quad \log \beta = L - T.$$

Hence

$$\alpha^p = e^{p(L+T)}, \quad \beta^q = e^{q(L-T)}.$$

The roots  $m\alpha^p, n\beta^q$  therefore have sum

$$me^{p(L+T)} + ne^{q(L-T)}$$

and product

$$mne^{p(L+T)+q(L-T)} = mne^{(p+q)L+(p-q)T}.$$

Thus the first quadratic follows.

Similarly,

$$\alpha^q = e^{q(L+T)}, \quad \beta^p = e^{p(L-T)}.$$

So the roots  $n\alpha^q, m\beta^p$  have sum

$$ne^{q(L+T)} + me^{p(L-T)}$$

and product

$$mne^{q(L+T)+p(L-T)} = mne^{(p+q)L+(q-p)T}.$$

This gives the second quadratic. □

## 7 Coefficient Form with L and T Notation

Substituting

$$S = -\frac{b}{a}, \quad P = \frac{c}{a}, \quad R = \sqrt{\left(\frac{b}{a}\right)^2 - 4\frac{c}{a}},$$

we obtain

$$L = \frac{1}{2} \log \left( \frac{c}{a} \right),$$

and

$$T = \frac{1}{2} \log \left( \frac{-\frac{b}{a} + \sqrt{\left(\frac{b}{a}\right)^2 - 4\frac{c}{a}}}{-\frac{b}{a} - \sqrt{\left(\frac{b}{a}\right)^2 - 4\frac{c}{a}}} \right).$$

Therefore the first transformed quadratic is

$$x^2 - \left[ me^{p(L+T)} + ne^{q(L-T)} \right] x + mne^{(p+q)L+(p-q)T} = 0.$$

The second transformed quadratic is

$$x^2 - \left[ ne^{q(L+T)} + me^{p(L-T)} \right] x + mne^{(p+q)L+(q-p)T} = 0.$$

For readability, the compact  $L, T$  notation is usually preferable.

## 8 Alternative Direct Root Form

Since

$$\alpha = \frac{S+R}{2}, \quad \beta = \frac{S-R}{2},$$

one may write the first transformed quadratic as

$$x^2 - \left[ m \left( \frac{S+R}{2} \right)^p + n \left( \frac{S-R}{2} \right)^q \right] x + mn \left( \frac{S+R}{2} \right)^p \left( \frac{S-R}{2} \right)^q = 0.$$

Similarly, the second transformed quadratic is

$$x^2 - \left[ n \left( \frac{S+R}{2} \right)^q + m \left( \frac{S-R}{2} \right)^p \right] x + mn \left( \frac{S+R}{2} \right)^q \left( \frac{S-R}{2} \right)^p = 0.$$

These forms are equivalent to the logarithmic expressions, but the logarithmic form is often numerically preferable when  $p, q$  are large or non-integer.

## 9 Special Cases

### 9.1 Equal scaling: $m = n$

If  $m = n = k$ , then the first equation becomes

$$x^2 - k \left( e^{p(L+T)} + e^{q(L-T)} \right) x + k^2 e^{(p+q)L+(p-q)T} = 0.$$

If also  $p = q$ , then the roots are

$$k\alpha^p, \quad k\beta^p.$$

The equation becomes

$$x^2 - k(\alpha^p + \beta^p)x + k^2(\alpha\beta)^p = 0.$$

## 9.2 Equal powers: $p = q$

If  $p = q = r$ , then the two transformed quadratics are

$$x^2 - (m\alpha^r + n\beta^r)x + mn(\alpha\beta)^r = 0,$$

and

$$x^2 - (n\alpha^r + m\beta^r)x + mn(\alpha\beta)^r = 0.$$

## 9.3 No power transformation: $p = q = 1$

If  $p = q = 1$ , then the main theorem reduces to Stage 1:

$$x^2 - (m\alpha + n\beta)x + mn\alpha\beta = 0.$$

## 9.4 No scaling: $m = n = 1$

If  $m = n = 1$ , then the first equation has roots

$$\alpha^p, \quad \beta^q,$$

and the second has roots

$$\alpha^q, \quad \beta^p.$$

# 10 Algorithmic Construction

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**Algorithm 1** Construct the two transformed quadratics

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**Require:** Coefficients  $a, b, c$  with positive distinct roots; parameters  $m, n, p, q$

**Ensure:** Coefficients of two monic quadratics

- 1:  $S \leftarrow -b/a$
  - 2:  $P \leftarrow c/a$
  - 3:  $R \leftarrow \sqrt{S^2 - 4P}$
  - 4:  $L \leftarrow \frac{1}{2} \log(P)$
  - 5:  $T \leftarrow \frac{1}{2} \log((S + R)/(S - R))$
  - 6:  $A_1 \leftarrow 1$
  - 7:  $B_1 \leftarrow -(me^{p(L+T)} + ne^{q(L-T)})$
  - 8:  $C_1 \leftarrow mne^{(p+q)L+(p-q)T}$
  - 9:  $A_2 \leftarrow 1$
  - 10:  $B_2 \leftarrow -(ne^{q(L+T)} + me^{p(L-T)})$
  - 11:  $C_2 \leftarrow mne^{(p+q)L+(q-p)T}$
  - 12: **return**  $(A_1, B_1, C_1), (A_2, B_2, C_2)$
- 

| Step                         | Operation                        | Complexity |
|------------------------------|----------------------------------|------------|
| Compute $S, P$               | two divisions                    | $O(1)$     |
| Compute $R$                  | square root                      | $O(1)$     |
| Compute $L, T$               | logarithms                       | $O(1)$     |
| Compute transformed sums     | exponentials and additions       | $O(1)$     |
| Compute transformed products | exponentials and multiplications | $O(1)$     |

Table 2: Computational complexity of the closed-form transformation.

# 11 Numerical Stability

The formula

$$R = \sqrt{S^2 - 4P}$$



can suffer from cancellation when  $S^2$  is close to  $4P$ . This corresponds to nearly equal roots.

The logarithmic coordinate

$$T = \frac{1}{2} \log \left( \frac{S+R}{S-R} \right)$$

also becomes small when  $\alpha \approx \beta$ . In practice, one may compute

$$T = \operatorname{arctanh} \left( \frac{R}{S} \right),$$

because

$$\frac{1}{2} \log \left( \frac{S+R}{S-R} \right) = \frac{1}{2} \log \left( \frac{1+R/S}{1-R/S} \right) = \operatorname{arctanh} \left( \frac{R}{S} \right).$$

This is often more stable when  $R/S$  is small.

| Quantity   | Direct formula                  | Stable alternative                 |
|------------|---------------------------------|------------------------------------|
| $R$        | $\sqrt{S^2 - 4P}$               | use high precision near $S^2 = 4P$ |
| $T$        | $\frac{1}{2} \log((S+R)/(S-R))$ | $\operatorname{arctanh}(R/S)$      |
| $\alpha^p$ | $((S+R)/2)^p$                   | $e^{p(L+T)}$                       |
| $\beta^q$  | $((S-R)/2)^q$                   | $e^{q(L-T)}$                       |

Table 3: Numerically stable forms of the core transformations.

## 12 Statistical Interpretation

Suppose the coefficients are estimated from data:

$$\hat{\theta} = (\hat{a}, \hat{b}, \hat{c})^\top.$$

Assume that for sample size  $N$ ,

$$\sqrt{N}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}(0, \Sigma).$$

Let  $F$  denote the transformation mapping

$$(a, b, c, m, n, p, q)$$

to the coefficient vector of the first transformed monic quadratic:

$$F_1(a, b, c; m, n, p, q) = \begin{pmatrix} 1 \\ - (me^{p(L+T)} + ne^{q(L-T)}) \\ mne^{(p+q)L+(p-q)T} \end{pmatrix}.$$

Similarly, define

$$F_2(a, b, c; m, n, p, q) = \begin{pmatrix} 1 \\ - (ne^{q(L+T)} + me^{p(L-T)}) \\ mne^{(p+q)L+(q-p)T} \end{pmatrix}.$$

By the multivariate delta method,

$$\sqrt{N} \left( F_j(\hat{\theta}) - F_j(\theta) \right) \xrightarrow{d} \mathcal{N} \left( 0, J_j \Sigma J_j^\top \right),$$

where

$$J_j = \frac{\partial F_j}{\partial (a, b, c)}$$

is the Jacobian of  $F_j$  with respect to the original quadratic coefficients.

This provides a standard asymptotic uncertainty estimate for the transformed coefficients.

## 12.1 Empirical Validation

Given observed estimates

$$\hat{a}, \hat{b}, \hat{c},$$

one may compute transformed coefficient estimates and validate them by checking whether the proposed roots satisfy the constructed quadratics.

For the first transformed equation, define

$$Q_1^*(x) = x^2 - \left( me^{p(L+T)} + ne^{q(L-T)} \right) x + mne^{(p+q)L+(p-q)T}.$$

A numerical residual score is

$$\mathcal{R}_1 = |Q_1^*(m\alpha^p)| + |Q_1^*(n\beta^q)|.$$

For exact symbolic arithmetic,

$$\mathcal{R}_1 = 0.$$

For floating-point arithmetic,  $\mathcal{R}_1$  should be close to machine precision.

---

**Algorithm 2** Monte Carlo validation of transformed quadratics

---

**Require:** Number of simulations  $N$ ; parameter ranges for  $a, b, c, m, n, p, q$

**Ensure:** Empirical residual distribution

- 1: **for**  $j = 1, \dots, N$  **do**
- 2:   Sample roots  $\alpha_j > \beta_j > 0$
- 3:   Construct  $a_j, b_j, c_j$  from chosen leading coefficient  $a_j$ :

$$b_j = -a_j(\alpha_j + \beta_j), \quad c_j = a_j\alpha_j\beta_j$$

- 4:   Sample  $m_j, n_j, p_j, q_j$
  - 5:   Construct transformed quadratics using Algorithm 1
  - 6:   Evaluate residuals at  $m_j\alpha_j^{p_j}, n_j\beta_j^{q_j}$
  - 7: **end for**
  - 8: Report mean, median, maximum, and quantiles of residuals
- 

## 13 Data Science and AI/ML Interpretation

The mapping

$$(a, b, c) \mapsto (L, T)$$

can be viewed as a feature map from the coefficient space of positive-root quadratics into a two-dimensional latent representation:

$$\Phi(a, b, c) = (L, T).$$

The inverse root map is

$$\alpha = e^{L+T}, \quad \beta = e^{L-T}.$$

The transformed roots are

$$me^{p(L+T)}, \quad ne^{q(L-T)}.$$

Thus, the full transformation is differentiable on the open domain

$$\mathcal{D} = \left\{ (a, b, c) : a \neq 0, \Delta > 0, -\frac{b}{a} > 0, \frac{c}{a} > 0 \right\}.$$

This differentiability allows the formula to be embedded in statistical learning systems as a deterministic layer.

### 13.1 Differentiable Layer

Define the root-power layer

$$\mathcal{L}_{m,n,p,q}(a,b,c) = (B_1, C_1, B_2, C_2),$$

where

$$B_1 = -\left(me^{p(L+T)} + ne^{q(L-T)}\right),$$

$$C_1 = mne^{(p+q)L+(p-q)T},$$

$$B_2 = -\left(ne^{q(L+T)} + me^{p(L-T)}\right),$$

and

$$C_2 = mne^{(p+q)L+(q-p)T}.$$

A neural network may predict unconstrained parameters  $\eta$ , which are then mapped into valid positive-root coefficients by a stabilising parametrisation, for example:

$$\alpha = e^{u+v}, \quad \beta = e^{u-v}, \quad v > 0,$$

with

$$a = 1, \quad b = -(\alpha + \beta), \quad c = \alpha\beta.$$

This guarantees

$$\alpha > \beta > 0$$

whenever  $v > 0$ .

### 13.2 Loss Function

If observed transformed coefficients are

$$(B_1^{\text{obs}}, C_1^{\text{obs}}, B_2^{\text{obs}}, C_2^{\text{obs}}),$$

a natural squared-error loss is

$$\mathcal{J} = (B_1 - B_1^{\text{obs}})^2 + (C_1 - C_1^{\text{obs}})^2 + (B_2 - B_2^{\text{obs}})^2 + (C_2 - C_2^{\text{obs}})^2.$$

One may add a discriminant regulariser

$$\lambda \max\{0, 4ac - b^2\}^2$$

to discourage invalid regions when learning directly in coefficient space.

## 14 Proof of Identifiability

**Lemma 1** (Identifiability of Positive Ordered Roots). *If*

$$S = \alpha + \beta, \quad P = \alpha\beta,$$

*with*

$$\alpha > \beta > 0,$$

*then  $\alpha, \beta$  are uniquely determined by  $S, P$ .*

*Proof.* The roots are solutions of

$$z^2 - Sz + P = 0.$$

The quadratic formula gives

$$z = \frac{S \pm \sqrt{S^2 - 4P}}{2}.$$

Since  $\alpha > \beta$ , the larger root is the plus branch and the smaller root is the minus branch:

$$\alpha = \frac{S + \sqrt{S^2 - 4P}}{2}, \quad \beta = \frac{S - \sqrt{S^2 - 4P}}{2}.$$

Therefore  $\alpha, \beta$  are uniquely determined. □

**Corollary 1** (Uniqueness of the Transformed Quadratics). *Under the assumptions of the main theorem, the two transformed monic quadratics are unique.*

*Proof.* The original coefficients determine  $\alpha, \beta$  uniquely. The parameters  $m, n, p, q$  then determine the transformed roots uniquely. Finally, a monic quadratic is uniquely determined by its root sum and root product.  $\square$

## 15 Worked Symbolic Example

Consider

$$x^2 - 5x + 6 = 0.$$

Then

$$\alpha = 3, \quad \beta = 2.$$

Here

$$S = 5, \quad P = 6, \quad R = 1.$$

Also,

$$L = \frac{1}{2} \log 6,$$

and

$$T = \frac{1}{2} \log \left( \frac{6}{4} \right) = \frac{1}{2} \log \left( \frac{3}{2} \right).$$

Let

$$m = 2, \quad n = 3, \quad p = 2, \quad q = 1.$$

The first transformed roots are

$$m\alpha^p = 2 \cdot 3^2 = 18,$$

and

$$n\beta^q = 3 \cdot 2 = 6.$$

Therefore the transformed quadratic is

$$x^2 - 24x + 108 = 0.$$

The second transformed roots are

$$n\alpha^q = 3 \cdot 3 = 9,$$

and

$$m\beta^p = 2 \cdot 2^2 = 8.$$

Therefore the second transformed quadratic is

$$x^2 - 17x + 72 = 0.$$

## 16 Discussion

The essential idea is that Viète's formulae provide the symmetric statistics

$$\alpha + \beta, \quad \alpha\beta,$$

while the ordering assumption  $\alpha > \beta$  selects the correct square-root branch. Positivity allows a logarithmic reparametrisation:

$$(\alpha, \beta) \longleftrightarrow (L, T),$$

where

$$L = \log \sqrt{\alpha\beta}, \quad T = \log \sqrt{\alpha/\beta}.$$

Power transformations are linear in logarithmic coordinates:

$$\log(\alpha^p) = p(L + T), \quad \log(\beta^q) = q(L - T).$$

Thus the apparently nonlinear root transformation

$$(\alpha, \beta) \mapsto (m\alpha^p, n\beta^q)$$

is decomposed into:

1. a coefficient-to-root map;
2. a root-to-log-root map;
3. a linear exponent scaling in log space;
4. a log-root-to-root map;
5. a root-to-polynomial reconstruction.

This decomposition is useful in symbolic algebra, numerical analysis, statistical modelling, and differentiable machine learning.

## 17 Conclusion

For a quadratic equation

$$ax^2 + bx + c = 0$$

with roots

$$\alpha > \beta > 0,$$

the two desired transformed quadratics are fully determined by

$$a, b, c, m, n, p, q.$$

With

$$\begin{aligned} S &= -\frac{b}{a}, & P &= \frac{c}{a}, & R &= \sqrt{S^2 - 4P}, \\ L &= \frac{1}{2} \log P, & T &= \frac{1}{2} \log \left( \frac{S+R}{S-R} \right), \end{aligned}$$

the two quadratics are

$$x^2 - \left( me^{p(L+T)} + ne^{q(L-T)} \right) x + mne^{(p+q)L+(p-q)T} = 0,$$

and

$$x^2 - \left( ne^{q(L+T)} + me^{p(L-T)} \right) x + mne^{(p+q)L+(q-p)T} = 0.$$

These formulas are exact, symbolic, computationally constant-time, and compatible with statistical and machine-learning workflows.

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## Glossary

**Quadratic equation** An equation of the form  $ax^2 + bx + c = 0$ , where  $a \neq 0$ .

**Root** A value  $r$  such that  $Q(r) = 0$ .

**Viète's formulae** Relations connecting polynomial coefficients with sums and products of roots.

**Discriminant** The quantity  $\Delta = b^2 - 4ac$ , determining the nature of the roots.

**Positive-root quadratic** A quadratic whose real roots are both positive.

**Monic quadratic** A quadratic whose leading coefficient is 1.

**Symmetric expression** An expression invariant under interchange of roots, such as  $\alpha + \beta$  or  $\alpha\beta$ .

**Ordered roots** Roots labelled so that  $\alpha > \beta$ .

**Logarithmic coordinate** A coordinate obtained by taking natural logarithms of positive variables.

**Mean log-root** The value  $L = \frac{1}{2}(\log \alpha + \log \beta)$ .

**Half log-ratio** The value  $T = \frac{1}{2}(\log \alpha - \log \beta)$ .

**Delta method** A statistical technique for approximating the asymptotic distribution of a smooth transformation of an estimator.

**Differentiable layer** A deterministic differentiable transformation used as a component of a machine-learning model.

## The End